

```
> restart;
> f := x^4 * sin(x);
```

$$f := x^4 \sin(x) \quad (1)$$

```
> sol := int(int(f, x=0..Pi), phi=0..1);
```

$$sol := 48 + \pi^4 - 12 \pi^2 \quad (2)$$

```
> midpoint_1 := N -> sum(sum(subs(x=i*Pi/N, f)*Pi/N, i=0..N)*1/(2*N), j=0..(2*N-1));
```

$$midpoint\_1 := N \rightarrow \sum_{j=0}^{2N-1} \frac{1}{2} \frac{\sum_{i=0}^N \frac{\text{subs}\left(x = \frac{i\pi}{N}, f\right) \pi}{N}}{N} \quad (3)$$

```
> midpoint_2 := N -> sum(sum(subs(x=i*Pi/N, f)*Pi/N, i=0..N)*1/(2*N), j=0..(2*N));
```

$$midpoint\_2 := N \rightarrow \sum_{j=0}^{2N} \frac{1}{2} \frac{\sum_{i=0}^N \frac{\text{subs}\left(x = \frac{i\pi}{N}, f\right) \pi}{N}}{N} \quad (4)$$

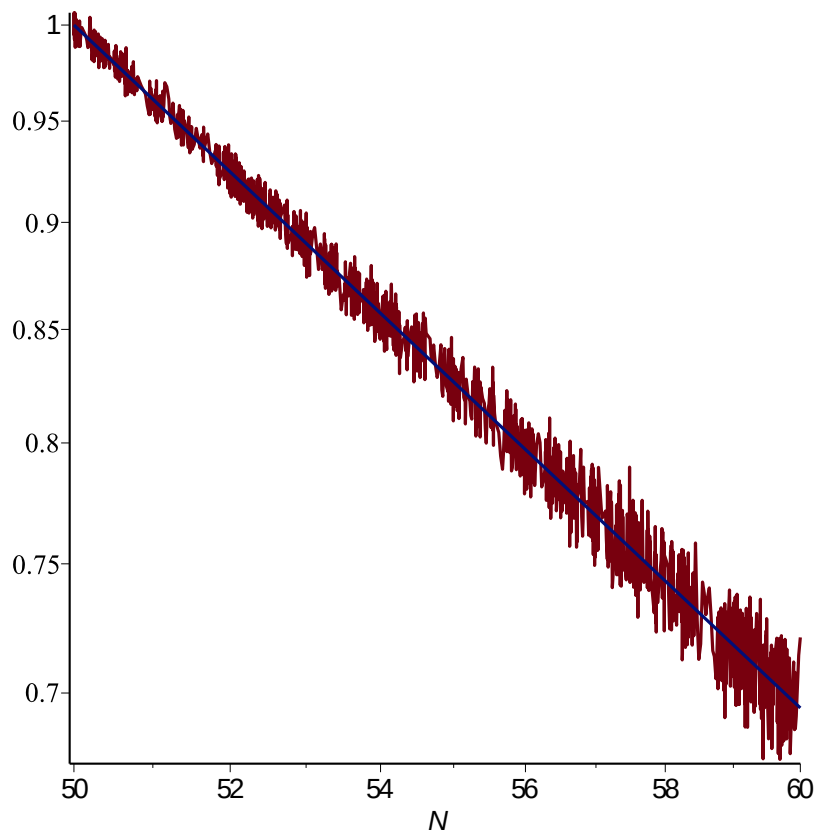
```
> abs_error_1 := N -> abs(midpoint_1(N) - sol);
```

$$abs\_error\_1 := N \rightarrow |midpoint\_1(N) - sol| \quad (5)$$

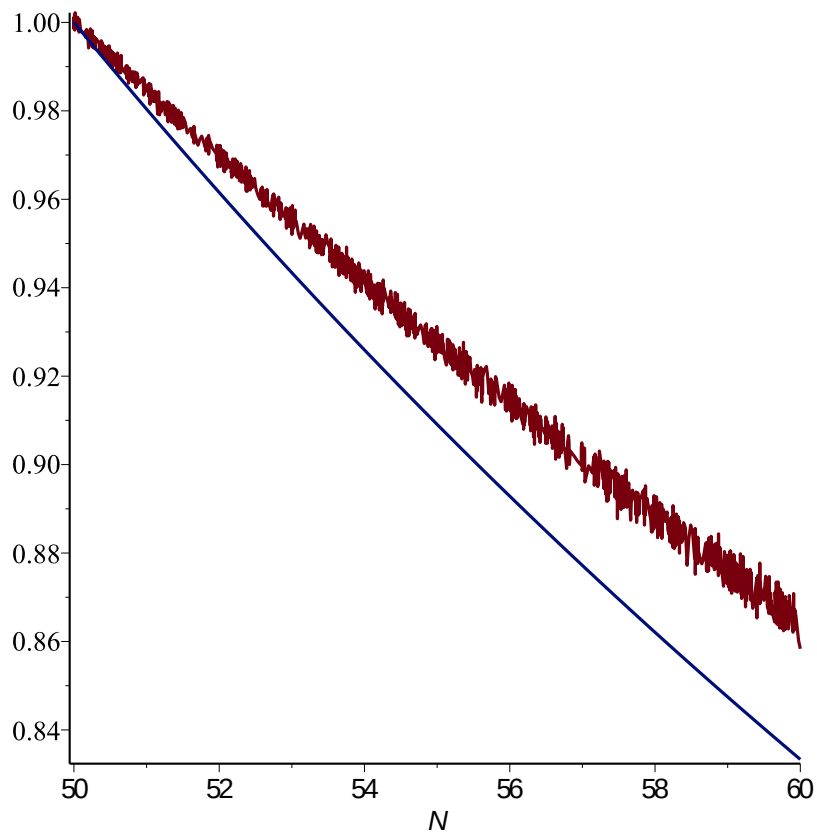
```
> abs_error_2 := N -> abs(midpoint_2(N) - sol);
```

$$abs\_error\_2 := N \rightarrow |midpoint\_2(N) - sol| \quad (6)$$

```
> plot(({abs_error_1(N)-abs_error_1(2*N))/(abs_error_1(50)-abs_error_1(2*50)), 50^2/N^2}, N=50..60);
```



```
> plot({(abs_error_2(N)-abs_error_2(2*N))/(abs_error_2(50)-  
abs_error_2(2*50)),50/N}, N=50..60);
```



```
> evalf(abs_error(100)); evalf(abs_error(200));  
      abs_error(100)  
      abs_error(200)
```

(7)